

MALMESBURY, South AfricaWMO: **687150**Lat: **33.472S**Lon: **18.718E**Elev: **331**StdP: **14.52**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.4	38.5	32.5	27.3	51.2	34.3	29.3	49.5	13.8	57.3	12.0	59.0	1.8	30	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.8	95.9	67.5	92.2	66.6	88.7	65.6	70.3	88.4	68.9	85.7	67.9	83.6	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	94.0	73.9	63.7	89.5	73.5	62.5	85.7	72.9	34.4	88.4	33.2	85.8	32.3	83.8	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.2	9.7	DB	33.7	104.9	1.4	3.0	32.7	107.1	31.9	108.8	31.2	110.5	30.2	112.7	(4)
(5)			WB	33.3	72.3	1.7	1.3	32.1	73.3	31.1	74.0	30.1	74.8	28.9	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	74.3	74.8	71.1	65.8	59.9	54.5	53.4	53.4	57.2	63.1	67.1	71.4	(6)	
	DBStd	9.26	5.46	5.68	5.37	6.23	4.26	3.86	4.19	3.59	4.93	6.21	5.53	5.16	(7)	
	HDD50	30	0	0	0	0	0	6	13	9	2	0	0	0	(8)	
	HDD65	1682	2	2	9	62	169	316	359	361	242	112	41	7	(9)	
	CDD50	5055	752	693	653	475	306	140	120	114	217	407	513	665	(10)	
	CDD65	1230	289	275	197	88	10	0	1	0	7	53	104	206	(11)	
	CDH74	10310	2551	2263	1309	686	79	8	14	8	84	581	915	1812	(12)	
	CDH80	4185	1152	1020	471	252	10	0	2	0	22	212	332	712	(13)	
(14) Wind	WSAvg	3.6	4.4	4.0	3.5	2.9	3.3	3.3	3.2	3.5	3.3	3.5	4.2	4.3	(14)	
(15) Precipitation	PrecAvg	18.2	0.4	0.4	0.7	1.6	2.4	3.2	3.1	2.5	1.7	1.1	0.7	0.6	(15)	
	PrecMax	31.9	1.9	1.7	2.8	5.3	6.1	8.3	8.1	9.3	4.3	3.5	4.1	2.5	(16)	
	PrecMin	10.1	0.0	0.0	0.0	0.0	0.3	0.3	1.0	0.6	0.3	0.0	0.0	0.0	(17)	
	PrecStd	4.6	0.4	0.5	0.7	1.3	1.6	1.9	1.7	1.5	1.0	1.0	0.9	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.2	100.0	95.7	94.2	81.1	74.5	75.3	74.2	83.8	92.9	94.6	97.4	(19)	
		MCWB	69.8	67.8	67.5	66.0	60.7	58.6	58.5	58.9	62.4	64.5	64.8	67.2	(20)	
	2%	DB	95.4	94.5	89.7	86.7	75.4	68.7	69.3	67.5	75.4	85.9	88.1	91.6	(21)	
		MCWB	68.6	67.4	66.2	63.2	59.6	57.1	57.2	57.1	58.7	62.7	62.9	65.7	(22)	
	5%	DB	90.9	90.2	85.2	81.0	71.2	64.6	64.8	64.0	70.3	80.6	83.4	87.4	(23)	
		MCWB	67.4	67.0	64.9	62.2	59.0	55.6	55.6	55.4	57.6	61.5	62.5	65.0	(24)	
	10%	DB	86.3	86.1	81.1	76.6	67.6	61.6	61.3	60.9	66.4	75.5	79.0	83.6	(25)	
		MCWB	66.5	66.4	64.1	61.2	58.0	54.7	54.2	53.7	56.4	59.8	61.6	64.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	71.7	69.3	67.2	63.7	60.6	61.2	61.0	64.1	67.2	67.5	70.1	(27)	
		MCDB	91.6	90.5	87.7	85.9	74.2	68.8	71.3	70.3	77.2	86.2	85.1	88.1	(28)	
	2%	WB	70.8	70.0	68.0	65.6	61.8	58.5	58.4	58.5	60.9	64.9	65.7	68.3	(29)	
		MCDB	89.7	87.0	85.4	82.3	70.0	64.9	66.9	65.8	73.0	82.1	81.5	85.3	(30)	
	5%	WB	69.2	68.6	66.5	63.9	60.5	57.3	56.6	56.3	58.8	62.6	64.3	67.2	(31)	
		MCDB	86.1	84.2	80.9	76.5	67.1	62.3	63.0	62.5	68.1	77.1	79.0	82.9	(32)	
	10%	WB	67.9	67.5	65.2	62.6	59.3	56.1	55.0	54.6	57.2	60.7	63.0	65.8	(33)	
		MCDB	82.9	82.0	78.0	73.3	65.4	60.5	60.1	59.7	65.2	72.4	75.8	79.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	23.8	23.8	22.5	22.0	17.8	17.7	19.1	17.6	19.6	22.9	22.9	23.5	(35)	
		MCDBR	31.4	30.0	27.9	29.5	24.9	23.8	26.0	24.4	27.2	32.1	31.9	30.8	(36)	
		MCWBR	9.4	8.5	8.8	10.4	12.0	13.8	15.7	14.9	13.0	12.3	11.1	10.1	(37)	
	5% WB	MCDBR	26.9	24.9	24.8	25.6	19.9	18.6	22.4	21.5	23.5	28.2	27.3	26.2	(38)	
		MCWBR	8.5	8.2	8.8	10.0	10.7	11.7	14.2	13.7	12.2	11.7	10.2	9.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.332	0.333	0.328	0.315	0.311	0.305	0.307	0.326	0.335	0.344	0.329	0.333	(40)	
	taud		2.509	2.510	2.520	2.557	2.562	2.553	2.538	2.483	2.425	2.414	2.498	2.500	(41)	
	Ebn at Noon		319	312	303	289	273	266	272	281	295	305	318	320	(42)	
	Edh at Noon		36	35	32	27	24	23	24	29	35	38	36	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2656	2387	1932	1407	976	823	919	1152	1587	2086	2476	2672	(44)	
	RadStd		64	64	64	73	65	48	57	55	65	103	69	72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (13 neighbors)	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	(47)

Nomenclature: See separate page